

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
AI Practical

Practical – VII

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Class: TY.IT	Batch: 1
Date of Assignment: 07/08/2026	Date/Time of Submission:07/08/2026

4. Implement k-fold cross-validation on a classification model.

Code:

```
from sklearn.datasets import load_iris
from sklearn.model_selection import cross_val_score
from sklearn.neighbors import KNeighborsClassifier
iris = load_iris()
X = iris.data
y = iris.target
model = KNeighborsClassifier()
scores = cross_val_score(model, X, y, cv=5)
print("Accuracy scores for each fold:")
print(scores)
print("Average Accuracy:", round(scores.mean() * 100, 2), "%")
```

Output:

```
Accuracy scores for each fold:
[0.96666667 1.          0.93333333 0.96666667 1.          ]
Average Accuracy: 97.33 %
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```

Dataset link : <https://www.kaggle.com/datasets/heptapod/titanic>

Code:

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
data = pd.read_csv("train_and_test2.csv")
data = data.fillna(0)
X = data.drop("2urvived", axis=1)
y = data["2urvived"]
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.3,
    random_state=1
)
```

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```
model = KNeighborsClassifier(n_neighbors=5)
model.fit(X_train, y_train)
print("Training Accuracy :", model.score(X_train, y_train) * 100)
print("Testing Accuracy :", model.score(X_test, y_test) * 100)
prediction = model.predict([X_test.iloc[0]])
print("Predicted Class :", prediction[0])
print("Actual Class :", y_test.iloc[0])
print("Omkar Prakash Sawant T040")
```

Output:

```
Training Accuracy : 84.82532751091702
Testing Accuracy : 72.264631043257
Predicted Class : 1
Actual Class : 0
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c:\Users\itlab\AppData\Local\Programs\Python\Python314\Lib\site-packages\sklearn\utils\validation.py:2827: UserWarning:
warnings.warn(
```

5. Implement a Decision Tree classifier using the Iris or any relevant dataset.

Code:

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
iris = load_iris()
X = iris.data
y = iris.target
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.3,
    random_state=1
)
model = DecisionTreeClassifier()
model.fit(X_train, y_train)
print("Training Accuracy:", model.score(X_train, y_train) * 100)
print("Testing Accuracy:", model.score(X_test, y_test) * 100)
prediction = model.predict([X_test[0]])
print("Predicted Class:", prediction[0])
print("Omkar Prakash Sawant T040")
```

Output:

```
Training Accuracy: 100.0
Testing Accuracy: 95.55555555555556
Predicted Class: 0
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```

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Dataset link : <https://www.kaggle.com/datasets/heptapod/titanic>

Code:

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
data = pd.read_csv("train_and_test2.csv")
X = data.drop("Survived", axis=1)
y = data["Survived"]
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.3,
    random_state=1
)
model = DecisionTreeClassifier()
model.fit(X_train, y_train)
print("Training Accuracy :", model.score(X_train, y_train) * 100)
print("Testing Accuracy :", model.score(X_test, y_test) * 100)
prediction = model.predict([X_test.iloc[0]])
print("Predicted Class :", prediction[0])
print("Actual Class   :", y_test.iloc[0])
print("Omkar Prakash Sawant T040")
```

Output:

```
Training Accuracy : 84.82532751091702
Testing Accuracy  : 72.264631043257
Predicted Class  : 1
Actual Class     : 0
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c:\Users\itlab\AppData\Local\Programs\Python\Python314\Lib\site-packages\sklearn\utils\validation.py:2827: UserWarning:
warnings.warn(
```